

Mahindra University Hyderabad
École Centrale School of Engineering
Minor-I exam

Program: B. Tech.

Branch: CM

Year: II

Semester: II

Subject: Foundational Mathematics for ML (MA2221)

Date: 26/02/2026

Start Time: 10:00 AM

Time Duration: 1.5 Hours

Max. Marks: 20

Instructions:

- 1) Answer all the questions.
- 2) All questions are self-explanatory; no clarification will be provided during the exam.
- 3) Use of one non-programmable scientific calculator is allowed.

1. (4 points) Answer the following in one or two lines.

- (a) State one application of the Singular Value Decomposition (SVD).
- (b) Why do we need to define the Moore–Penrose inverse?
- (c) State one use of projection in linear algebra.
- (d) Why is the matrix representation of a linear transformation needed?

2. (5 points) Let

$$\Phi : \mathbb{R}^3 \rightarrow \mathbb{R}^4$$

be a linear mapping defined by

$$\Phi \left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \right) = \begin{bmatrix} 2x_1 - x_2 + x_3 \\ x_1 + 3x_2 \\ -x_1 + x_3 \\ x_1 + 2x_2 + x_3 \end{bmatrix}.$$

Let

$$\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \right\} \text{ be a basis of } \mathbb{R}^3,$$

and

$$\mathcal{C} = \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} \right\} \text{ be the standard basis of } \mathbb{R}^4.$$

Find the matrix representation A_Φ of the linear transformation Φ with respect to the bases $\mathcal{B}$ and $\mathcal{C}$.

3. (3 points) Compute the angle between

$$x = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad y = \begin{bmatrix} -1 \\ -1 \end{bmatrix}$$

using

$$\langle x, y \rangle = x^T B y$$

where

$$B = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}.$$

4. (3 points) Use Gram-Schmidt process to find an orthonormal basis of

$$\text{span} \left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ 2 \\ 0 \end{bmatrix} \right\}.$$

5. (5 points) Find the singular value decomposition (SVD) of

$$A = \begin{bmatrix} 6 & 2 \\ 2 & 6 \end{bmatrix}.$$